Analytics competitors configure business rules employing quantitative tools, such as the techniques shown earlier.* The result of BPMSs incorporating additional analytics tools (e.g., data mining) as well as business rules management functionality is that the quality of the decisions in automated business processes tends to increase.

11.2.3 Monitor and Control

The lifecycle of business process excellence consists of the design, implementation, execution, and monitoring and control of processes (Kirchmer 2011). The business process reengineering movement that started in the 1990s emphasized the designed phase of the process lifecycle. It wasn't until a decade later that the application of technology and quantitative analysis became an important part of monitoring and controlling the execution of business processes. As mentioned in Section 11.2, BPMSs record many types of events that occur during process executions, including the start and completion time of each activity, input and output data, resource involved in the completion of activities, and even failures. The monitor and control functionality of BPMSs is made possible by the addition of *process intelligence* (PI), which is defined as the set of integrated tools that support business and IT analysts in managing process execution quality (Grigori et al. 2004). These tools are standard in most modern BPMSs.

Process logs (also called *event logs*) are the main input to PI tools. PI tools are designed to extract knowledge from process logs stored in data warehouses, where the logs have been cleaned and aggregated. The discovery might lead to identifying high- or low-quality process executions to find explanations of why they occurred as well as predicting future behavior. Data visualization tools play an important role in monitor and control activities. Dashboards and mashups are two common process intelligence tools related to these activities. *Dashboards* are visual displays that show the status or "health" of a business process via numeric and graphical key performance indicators.† *Mashups* refer to the integration of information from different sources by using editors that allow the remix of data without programming. Mashup in music means blending existing content to create something new. The technology behind mashups enables analysts to combine and aggregate process information from multiple sources to create and manage their own dashboards.

Business activity monitoring (BAM), an additional PI tool, is designed to determine abnormal events in the process and issue alerts when predefined conditions occur. In other words, BAM is software for the real-time monitoring of business processes. An example of a BAM alert is a credit card that has been charged multiple times in a short period of time. This might trigger an alert to the fraud prevention department. Another example of a BAM alert occurs when inventory of a particular product is below a specified level at retail stores. This alert might trigger an automatic process to place a purchase order with the vendor to delivery merchandise at specified points of the supply chain. The implementation of BAM does not necessarily have to involve BPMSs. However, BPMS engines and the business process designer contain the design- and run-time tools that facilitate the implementation of BAM solutions.

* The combination of rules and analytics is sometimes referred to as *enterprise decision management*.

† Financial or nonfinancial metrics that quantify business process performance, for example, cycle time.

TABLE 11.8
Process log

Case	Activity
1	a
2	a
3	a
3	b
1	b
1	c
2	c
4	a
2	b
2	d
5	e
4	c
1	d
3	c
3	d
4	b
5	f
4	d

11.2.4 Process Mining

Process mining is a set of techniques that are useful to analyze existing processes based on how they are actually being executed. Process mining operates on process logs that contain data associated with the order in which events took place. This is why these analytics tools are embedded in BMPSs. There are three classes of process mining techniques:

- *Discovery.* A set of procedures designed to construct a model of a process from data associated with the actual execution of the process.* In other words, without an *a priori* model, information is gathered about the process as it takes place, and an explicit process model is extracted. The model of the actual process should be consistent with the observed dynamic behavior.
- *Conformance analysis.* In this case, there is a process model that is compared with the event log, and discrepancies between the log and the model are analyzed.
- *Enhancement.* A discovered or existing model is enhanced by the analysis of information related to performance, cost, or social structures.

The principles of discovery—the main process mining technique—may be illustrated with the following example from van der Aalst and Weijters (2004). Consider the process log in Table 11.8. This log contains information about five cases (i.e., process instances). The log shows that Tasks *a*, *b*, *c*, and *d* have been executed for Cases 1 to 4. For Case 5, only two different tasks (*e* and *f*) were executed. This log can be represented as a set of traces or sequences in the following form:

* Automated process discovery is another term used to describe these process mining techniques.

$$L = \left[\langle a,b,c,d\rangle^2, \langle a,c,b,d\rangle^2, \langle e,f\rangle\right]$$

Cases 1 and 3 follow the first sequence; hence the superscript two. Cases 2 and 4 follow the second sequence. Case 5 is represented by the last sequence. A simple event log L is the main input to process discovery algorithms, which are defined as functions that map L onto a workflow network (WF-net). The α-algorithm is an example of a mapping function for process discovery (van der Aalst, 2016). This algorithm was one of the first process discovery algorithms to be able to handle concurrency. The basic α-algorithm is capable of discovering a large class of WF-nets, but it does have some limitations. In particular, it might generate WF-nets that are unnecessarily complex. In addition, it does not handle short loops (i.e., loops of length one or two) correctly. Some extensions to the original algorithm have been created that are able to correct these problems.

The quality of the model produced by a process discovery algorithm depends not only on the algorithm but also on the completeness of the event log. It cannot be assumed that the event log will contain all traces. Therefore, the model must be able to extrapolate. The issues of overfitting and underfitting that are typical of any data mining technique are also present in process discovery. Therefore, an event log is typically split into a training log and a test log. The training log is used to discover the process model, and the test log is used to evaluate the model. Van der Aalst (2016) proposes the following four dimensions to measure the quality of a model resulting from a process discovery:

- *Fitness.* This refers to the ability of the model to replay the traces in the event log. A model has a perfect fitness if it is capable of replaying all traces in the log.
- *Simplicity.* The simplest model that can explain the behavior in the log is considered the best. A measure of simplicity is the number of nodes and arcs in the resulting process graph.
- *Generalization.* Since it is not possible for a log to contain all possible traces, a model must be able to generalize. An overfitted model is one that is not able to generalize.
- *Precision.* A precise model does not allow behavior that is not possible. An underfitted model results in over-generalization; that is, it allows more behavior when there are no indications in the log that suggest this additional behavior. A model that is not precise allows behavior that is very different from what can be observed in the log.

The α-algorithm is based on establishing the relationships between all the activities in a log L:

- $a \rightarrow b$: indicates that b immediately follows a, and a never immediately follows b
- $a \# b$: indicates that neither b ever immediately follows a nor a ever immediately follows b
- $a \| b$: indicates that both b immediately follows a and a immediately follows b

Because precisely one of these relations holds for any pair of activities, a footprint of a log L can be captured in a matrix. Table 11.9 shows the footprint for the event log in Table 11.8. If Task b is executed, then Task c is also executed. However, Task c does not always follow Task b. For example, in the second trace of the log L above, Task c is executed before Task b. Based on the information shown in Table 11.8 and summarized

TABLE 11.9

Footprint for log in Table 11.8

	a	*b*	*c*	*d*	*e*	*f*
a		→	→	#	#	#
b			‖	→	#	#
c				→	#	#
d					#	#
e						→

as log *L*, with the corresponding footprint shown in Table 11.9, the process model in Figure 11.1 may be constructed. It is assumed that the cases are representative and the set of observed events is sufficiently large to characterize the behaviors of the process. The model includes an arc for each pair of activities labeled with an arrow (→) in the footprint of Table 11.9. Activities *b* and *c* are shown in parallel, as indicated in the footprint (||). Note that *b* and *c* appear in both orders, immediately following each other in the event log. The process starts with either Task *a* or Task *e*. Task *f* always follows Task *e*, and Task *d* always follows Tasks *b* and *c*.

Table 11.8 contains the minimum information that is required to extract a process model. However, process logs generated by BPMSs often include timestamps and case attributes that can be used to extract additional causality information. For this example, process discovery is relatively simple. However, large processes represent a challenge because of the large number of possible combinations generated by the presence of alternative and parallel routes. In addition, paths with low probability of occurring are difficult to detect. Extensions of the α algorithm have been suggested to remedy some of its shortcomings. For instance, the *heuristics miner* takes into consideration the number of times that each relationship between activities occurs (e.g., the number of times that *a* immediately followed *b* in the event log). These counts are used to build a *dependency graph* that shows a numerical relationship between pairs of events. The larger the values, the stronger the dependency. The heuristics are designed to find reliable causality relations even if the event log contains noise.

The α algorithm and the heuristics miner are just two of a variety of process mining procedures. A wealth of information about process mining, which is beyond the scope of this book, can be found in www.processmining.org. This website includes process-mining tools embedded in ProM, the process-mining framework, as well as information associated with Will van der Aalst's seminal book on process mining, titled *Process Mining: Data Science in Action*.

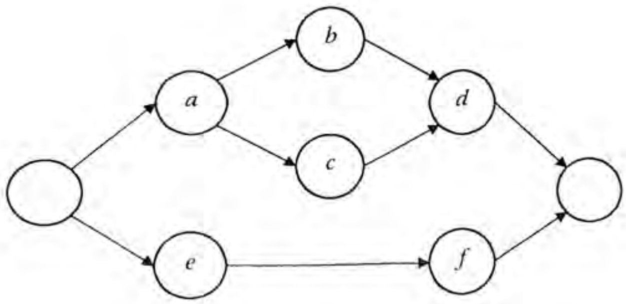


FIGURE 11.1

Process model discovered from the event log in Table 11.8.

11.3 Process Benchmarking

Process benchmarking has been defined as the systematic search for best-practice organizations with which to compare key processes to learn from observations and incorporate what these processes do well, with the goal of improving performance. The planning stage is critical in process benchmarking, because it consists of identifying the best-practice organizations that will serve as the basis for learning. *Data envelopment analysis* (DEA) is a predictive analytics tool that can be used for identifying best practices.

DEA, introduced by Charnes et al. (1978), is a linear programming method for calculating relative efficiencies of a set of organizations that possess some common functional elements but whose efficiency may vary due to internal differences. One main difference, for instance, might be the management style employed in each organization. In DEA, organizations are referred to as *decision-making units* (DMUs).

DEA has generated a fair amount of interest in the academic world and among practitioners, because it has been applied successfully to assess the efficiency of various organizations in the public and private sectors. The popularity of the technique is evident from the increasing number of articles in scientific and practitioner journals. A search of DEA on the Internet results in hundreds of pages that make reference to this methodology. Recently, DEA has been used as a tool for benchmarking. This section discusses the application of DEA to business process benchmarking. When DEA is used for benchmarking processes, the process becomes the decision-making unit. Therefore, the terms *DMU* and *process* are used interchangeably throughout this section.

DEA considers a process as a black box and analyzes inputs and outputs to determine relative efficiencies. The black-box model (or transformation model) is depicted in Figure 11.2, where possible inputs to and outputs from a process are as follows:

Inputs	Outputs
• Full-time equivalent (FTE) employees	• Cycle time
• Office space	• Cycle time efficiency
• Regular hours	• Throughput rate
• Overtime hours	• Capacity utilization
• Operating expenses	• Customer rating
• Equipment (e.g., computers, telephone lines)	• On-time deliveries

Generally speaking, inputs are resources used by the process to produce outputs, which can be measured with key performance indicators. Using the black-box approach, the efficiency of a process can be calculated with a simple ratio when there is a single input and a single output.

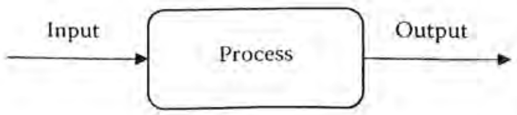


FIGURE 11.2
Black-box view of a process.

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}}$$

However, when a process uses multiple inputs and produces multiple outputs, it becomes more difficult to evaluate its efficiency.

Example 11.4: Efficiency Calculation

Suppose that labor cost and throughput rate are considered a single input and output of a business process, respectively. The efficiency of two processes, say A and B, can be obtained easily as shown in Table 11.10. Clearly, Process A is more efficient than Process B, because A is able to complete 0.75 jobs for every dollar spent in labor, but B can complete only 0.733 jobs per dollar. However, the manager of Process B might not agree with this assessment, because that manager might argue that efficiency is more directly affected by office space than it is by direct labor costs. The efficiency assessment in this case might be as shown in Table 11.11.

The new assessment shows that Process B is more efficient than Process A when office space is considered as the single input to this process. More realistically, both inputs should be used to compare the efficiency of Processes A and B.

The DEA approach offers a variety of models in which multiple inputs and outputs can be used to compare the efficiency of two or more processes. This section is limited in scope to the ratio model (also known as the Charnes–Cooper–Rhodes or CCR model), which is based on the following definition of efficiency:

$$\text{Efficiency} = \frac{\text{Weighted sum of outputs}}{\text{Weighted sum of inputs}}$$

Suppose this definition is used to compare the efficiency of the processes in Example 11.4 by considering that the labor cost is twice as important as the cost of office space. In this case, the efficiency of each process can be calculated as follows:

$$\text{Efficiency}\left(\text{Process A}\right) = \frac{1500}{2000 \times 2 + 10{,}000 \times 1} = 0.107$$

$$\text{Efficiency}\left(\text{Process B}\right) = \frac{1100}{1500 \times 2 + 6900 \times 1} = 0.111$$

TABLE 11.10

Efficiency calculation based on direct labor costs

Process	Labor Cost ($/week)	Throughput (jobs/week)	Efficiency (jobs/$)
A	$2000	1500	0.750
B	$1500	1100	0.733

TABLE 11.11

Efficiency calculation based on office space

Process	Office Area (ft^2)	Throughput (jobs/week)	Efficiency (jobs/ft^2)
A	10,000	1,500	0.15
B	6,900	1,100	0.16

The chosen weights make Process B more efficient than Process A, but the manager of Process A could certainly argue that a different set of weights might change the outcome of the analysis. The manager's argument is valid, because the weights were chosen arbitrarily. Even if the weights were chosen prudently, it would be almost impossible to build consensus about these values among all process owners. This is why DEA models are based on the premise that each process should be able to pick its own "best" set of weights. However, the weight values must satisfy the following conditions:

- All weights in the chosen set should be strictly greater than 0
- The set of weights cannot make any process in the comparison group more than 100 percent efficient

Suppose the owner of Process A in Example 11.4 chooses a weight of 0.25 for labor cost and 0.1 for office area. This set of weights makes Process A 100 percent efficient.

$$\text{Efficiency}\left(\text{Process A}\right)=\frac{1500}{2000\times 0.25+10{,}000\times 0.1}=1$$

However, the set of weights is not "legal" when used to compare the efficiency of Process A with the efficiency of Process B, because the weights result in an efficiency value for Process B that is more than 100 percent.

$$\text{Efficiency}\left(\text{Process B}\right)=\frac{1100}{1500\times 0.25+6900\times 0.1}=1.033$$

As seen in the previous example, the weight values can be "legal" or "illegal" depending on the processes that are being compared. Also, the efficiency of one process depends on the performance of the other processes that are included in the set. In other words, DEA is a tool for evaluating the relative efficiency of DMUs; therefore, no conclusions can be drawn regarding the absolute efficiency of a process when applying this technique.

Typical statistical methods are characterized as central tendency approaches, because they evaluate processes relative to the performance of an average process. In contrast, DEA is an extreme point method that compares each process with only the "best" processes. A fundamental assumption in an extreme point method is that if a given process A is capable of producing *Out*(A) units of output with *In*(A) units of input, then other processes also should be able to do the same if they were to operate efficiently. Similarly, if Process B is capable of producing *Out*(B) units of output with *In*(B) units of input, then other processes also should be capable of the same production efficiency. DEA is based on the premise that Processes A, B, and others can be combined to form a composite process with composite inputs and composite outputs. Because this composite process does not necessarily exist, it typically is called a *virtual process*.

The heart of the analysis lies in finding the best virtual process for each real process. If the virtual process is better than the real process, by either producing more output with the same input or producing the same output with less input, then the real process is declared inefficient. The procedure of finding the best virtual process is based on formulating a linear program. Hence, analyzing the efficiency of n processes consists of solving a set of n linear programming problems.

11.3.1 Graphical Analysis of the Ratio Model

Before the linear programming formulation of the ratio model is described, the ratio model will be examined for the case of a single input and two outputs. This special case can be studied and solved on a two-dimensional graph.

Example 11.5: Graphical Analysis of a Model with One Input and Two Outputs

Suppose that it is desired to compare the relative efficiency of six processes (labeled A to F) according to their use of a single input (labor hours) and two outputs (throughput rate and customer service rating). The relevant data values are shown in Table 11.12. (For the sake of this illustration, do not consider the relative magnitudes and/or the units of the data values.)

The ratio model can be used to find out which processes are relatively inefficient and also how the inefficient processes could become efficient. To answer these questions, first calculate two different ratios: (1) the ratio of throughput with respect to labor and (2) the ratio of customer rating with respect to labor. Table 11.13 shows the values of the independent efficiency ratios, where the first ratio is labeled x and the second ratio is labeled y. The x and y values from Table 11.13 are used to plot the relative position of each process in a two-dimensional coordinate system. The resulting graph is shown in Figure 11.3.

Figure 11.3 shows the efficient frontier of this benchmarking problem. Processes that lie on the efficient frontier are nondominated, but not all nondominated processes are efficient. A nondominated process is such that no other process can be at least as efficient in all the different performance measures and strictly more efficient in at least one performance measure. The nondominated processes that are in the outer envelope of the graph define the efficient frontier. Under this definition, Processes A, B, and C characterize the efficient frontier in Figure 11.3. These are also called the *relatively efficient processes*. Note that A is better than B and C in terms of the y-ratio, but it is inferior to both of those processes in terms of the x-ratio. Also note that E is a nondominated process that is not efficient, because it does not lie on the efficient frontier.

This graphical analysis has helped answer the question about the relative efficiency of the processes. A, B, and C are relatively efficient processes, and D, E, and F are relatively inefficient. The term *relatively efficient* or *inefficient* is used because the efficiency depends on the processes that are used to perform the analysis. To clarify this issue, suppose a process G is added with $x = 2.5$ and $y = 1$. This addition makes Process B relatively inefficient, because the revised envelope moves farther up (i.e., in the direction of the positive y values) relative to the original processes. In the absence of such a process, B is relatively efficient.

TABLE 11.12

Input and output data for Example 11.5

Process	Labor	Throughput	Rating
A	10	10	10
B	15	30	12
C	12	36	6
D	22	25	16
E	14	31	8
F	18	27	7

TABLE 11.13

Independent efficiency calculations

Process	x = throughput/labor	y = customer rating/labor
A	1.000	1.000
B	2.000	0.800
C	3.000	0.500
D	1.136	0.727
E	2.214	0.571
F	1.500	0.389

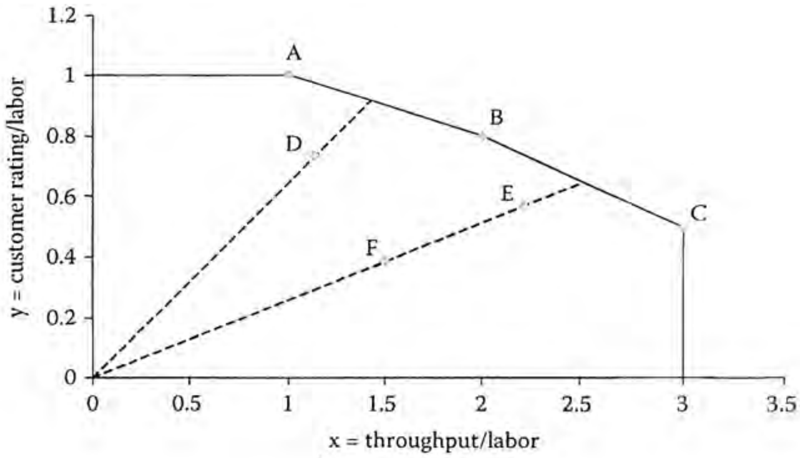


FIGURE 11.3
Efficient frontier for Example 11.5.

11.3.1.1 Efficiency Calculation

Relatively efficient processes (i.e., those on the efficient frontier) are considered to have 100 percent efficiency under the ratio model. What is the relative efficiency value of a process that is relatively inefficient? To answer this question, analyze the situation depicted in Figure 11.4. This figure shows a relatively inefficient process P0 with coordinates (x_0, y_0) and relatively efficient processes P1 and P2 with coordinates (x_1, y_1) and (x_2, y_2), where x = output 1/input and y = output 2/input. Because Processes P1 and P2 are relatively efficient, their efficiency value is 100 percent. Also, because the line going from the origin to Process P0 intersects the line from Process P1 to Process P2, these two processes are known as the *peer group* (or *reference set*) of Process P0.

The efficiency associated with Process P0 is less than 100 percent, because this process is not on the efficient frontier. The efficiency of Process P0 is the distance from the origin to the (x_0, y_0) point divided by the distance between the origin and the virtual process with coordinates (x_v, y_v). To calculate the efficiency of Process P0, it is necessary to first calculate the coordinates of the virtual (and efficient) process. This can be done using the following equations:

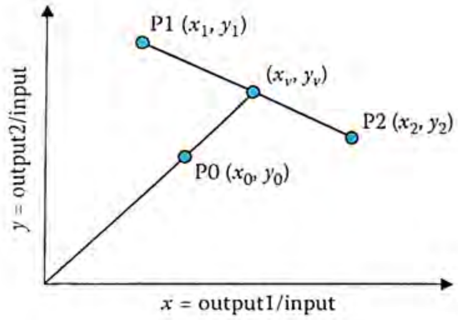


FIGURE 11.4
Projection of a relatively inefficient process on to the efficient frontier.

$$a = \frac{y_2 - y_1}{x_2 - x_1} \qquad b = \frac{x_2 y_1 - x_1 y_2}{x_2 - x_1}$$

$$x_v = \frac{b}{\frac{y_0}{x_0} - a} \qquad y_v = a x_v + b$$

Then, the efficiency of Process P0 is given by

$$E_0 = \sqrt{\frac{x_0^2 + y_0^2}{x_v^2 + y_v^2}}$$

To illustrate the use of these equations, consider Process D in Tables 11.12 and 11.13 and Figure 11.3. This process is relatively inefficient, and its peer group consists of Processes A and B. First, the coordinates of the virtual process corresponding to the real Process D can be calculated.

$$a = \frac{y_B - y_A}{x_B - x_A} = \frac{0.8 - 1}{2 - 1} = -0.2$$

$$b = \frac{x_B y_A - x_A y_B}{x_B - x_A} = \frac{2 \times 1 - 1 \times 0.8}{2 - 1} = 1.2$$

$$x_v = \frac{b}{\frac{y_D}{x_D} - a} = \frac{1.2}{\frac{0.727}{1.136} + 0.2} = 1.428$$

$$y_v = a x_v + b = -0.2 \times 1.428 + 1.2 = 0.914$$

The coordinates of the efficient virtual process allow the calculation of the relative efficiency of Process D.

$$E_0 = \sqrt{\frac{x_0^2 + y_0^2}{x_v^2 + y_v^2}} = \sqrt{\frac{1.136^2 + 0.727^2}{1.428^2 + 0.914^2}} = 0.795$$

The relative efficiency of Process D is 79.5 percent. Process D can become relatively efficient by moving toward the efficient frontier. The movement does not have to be along the line defined by the current position of Process D and the origin; in other words, Process D does not have to become the virtual process that was defined to measure its relative efficiency. Because Process D can become efficient by moving toward the efficiency frontier, Process D can become efficient in an infinite number of ways. These multiple possibilities involve a combination of using less input to produce more output and producing more output with the same input.

This analysis shows that DEA is not only able to identify relatively inefficient units, but it also is capable of setting targets for these units so that they become relatively efficient. When used in the context of benchmarking processes, DEA identifies the best-practice processes and also gives the inefficient processes numerical targets to achieve efficiency relative to their peers.

A set of targets for Process D in Example 11.5 can be calculated, based first on fixing the labor hours and then fixing the throughput and customer ratings. If the labor hours remain fixed, then Process D must increase its output to the following values to move to the efficient frontier:

$$\text{Throughput} = x_v \times \text{labor} = 1.428 \times 22 = 31.4$$

$$\text{Customer rating} = y_v \times \text{labor} = 0.914 \times 22 = 20.1$$

This means that if throughput is increased from 25 to 31.4, customer ratings are increased from 16 to 20.1, and the labor hours remain at 22, Process D becomes relatively efficient by moving to the coordinates of its corresponding virtual process on the efficient frontier. The process also can move to the coordinates of the virtual process by using fewer resources to produce the same output. The new input can be calculated as follows:

$$\text{Labor} = \frac{\text{Throughput} + \text{Customer rating}}{x_v + y_v} = \frac{25 + 16}{1.428 + 0.914} = 17.5$$

Therefore, Process D becomes relatively efficient if it reduces its input from 22 to 17.5.

11.3.2 Linear Programming Formulation of the Ratio Model

The ratio model introduced in the previous section is based on the idea of measuring the efficiency of a process by comparing it with a hypothetical process that is a weighted linear combination of other processes. The model measures relative efficiency as the weighted sum of outputs divided by the weighted sum of inputs.

The main assumption in the previous illustrations was that this measure of efficiency requires a common set of weights to be applied across all processes. This assumption immediately raises the question of how such an agreed-on common set of weights can be obtained. Two kinds of difficulties can arise in obtaining a common set of weights. First, it might simply be difficult to value the inputs or outputs. For example, different processes might choose to organize their operations differently, so that the relative values of the different outputs are legitimately different. This perhaps becomes clearer if one considers an attempt to compare the relative efficiency of schools with achievements in music and sports among the outputs. Some schools might legitimately value achievements in sports or music differently from other schools. Therefore, a measure of efficiency that requires a single common set of weights is unsatisfactory.

The DEA model recognizes the legitimacy of the argument that processes might value inputs and outputs differently and therefore, adopts different weights to measure efficiency. The model allows each process to adopt a set of weights that shows it in the most favorable light in comparison with the other processes.

The DEA ratio model, in particular, is formulated as a sequence of linear programs (one for each process) with the following characteristics:

Maximize the efficiency of one process

Subject to the efficiency of all processes ≤ 1

The variables in the model are the weights assigned to each input and output. A linear programming formulation of the ratio model that finds the best set of weights for a given process p among n processes with m inputs and q outputs is

$$\text{Maximize } \sum_{j=1}^{q} b_{jp} h_j$$

$$\text{Subject to } \sum_{i=1}^{m} a_{ip} g_i = 1$$

$$\sum_{j=1}^{q} b_{jk} h_j - \sum_{i=1}^{m} a_{ik} g_i \leq 0 \quad \text{for } k = 1, \ldots, n$$

$$g_i \geq 0.0001 \quad \text{for } i = 1, \ldots, m$$

$$h_j \geq 0.0001 \quad \text{for } j = 1, \ldots, q$$

The decision variables in this model are

g_i = the weight assigned to input i

h_j = the weight assigned to output j

Because there are m inputs and q outputs, the linear program consists of $m+q$ variables. The data are represented by the following:

a_{ik} = the amount of input i used by process k

b_{jk} = the amount of output j produced by process k

The linear programming model finds the set of weights that maximizes the weighted output for Process p. This maximization is subject to forcing the weighted input for Process p to be equal to 1. The efficiency of all units also is restricted to be less than or equal to 1. Therefore, Process p will be efficient if a set of weights is found such that the weighted output also is equal to 1. Because all weights must be strictly greater than 0 (that is, no input or output should be totally ignored), the last two sets of constraints in the model force the weights to be at least 0.0001.

Using the data in Table 11.12, the linear programming model can be formulated and used to calculate the relative efficiency of Process D.

Maximize $25h_1 + 16h_2$
Subject to $22g = 1$

$$10h_1 + 10h_2 - 10g \le 0$$

$$30h_1 + 12h_2 - 15g \le 0$$

$$36h_1 + 6h_2 - 12g \le 0$$

$$25h_1 + 16h_2 - 22g \le 0$$

$$31h_1 + 8h_2 - 14g \le 0$$

$$27h_1 + 7h_2 - 18g \le 0$$

$$h_1, h_2, g \ge 0.0001$$

In the formulation of the DEA model for Process D, this process is allowed to choose values for the weight variables that will make its efficiency calculation as large as possible. However, these values cannot be 0 and cannot make another process more than 100 percent efficient. The constraint that forces the efficiency of Process A to be less than or equal to 1

$$10h_1 + 10h_2 - 10g \le 0$$

is equivalent to

$$\frac{10h_1 + 10h_2}{10g} \le 1$$

However, a linear constraint is the standard form for formulating restrictions in a linear programming model, and this is why the ratio equation is transformed.

From solving this problem graphically, it is known that no values for h_1, h_2, and g can make Process *D* 100 percent efficient without violating at least one of the constraints. (If this is not convincing, give it a try.)

For benchmarking problems with processes that use several inputs and outputs, the DEA models typically are solved using specialized software. These packages provide a friendly way of capturing the problem data, and they automate the task of solving the linear programming problem for each process. Barr (2004) provides a survey of the best commercial and non-commercial DEA software tools. The survey includes descriptions of individual packages, comparisons of their features and capabilities, and links to further information. The ratio model discussed in this section is only one of many models associated with DEA. Barr's survey shows that collectively, the eight software packages that he selected as the best (four commercial and four non-commercial) include 24 different DEA models (including the ratio model). In the Appendix, a simple Excel add-in is described that performs DEA with the ratio model.

11.3.3 Learning from Best-Practice Organizations

One of the main benefits of DEA is that it identifies best-practice DMUs in a comparison set. For instance, Sherman and Ladino (1995) report their experiences with the application of DEA in a bank. The analysis resulted in a significant improvement in branch productivity and profits while maintaining service quality. The analysis identified more than $6 million of annual expense savings not identifiable with traditional financial and operating ratio analysis. The DEA models were used to compare branches objectively to identify the best-practice branches (those on the efficient frontier), the less productive branches, and the changes that the less productive branches needed to make to reach the best-practice level and to improve productivity.

The model used inputs such as number of customer service tellers, office square footage, and total expenses (excluding personnel and rent). The outputs that were considered included number of deposits, withdrawals, loans, new accounts, bank checks, and traveler's checks. Out of the 33 branches, the analysis identified that 10 were relatively efficient. The peer groups for the relatively inefficient branches were used to identify the operating characteristics that made the less productive branches more costly to operate.

The two main questions that benchmarking attempts to answer are: (1) Where are the best practices within a group of teams or regions? and (2) What are the best-practice organizations doing differently? The first question may be addressed with analytical tools such as DEA. The second question relates to the inner workings of a best-practice unit. This may require the examination of both process and organizational structure. Flowcharts—or the more advanced versions known as *event-driven process chains*—may be used to compare process structures and to identify differences between relatively efficient processes and those that are not.

Detailed analysis of the organizational structure may also provide valuable information related to the performance of individuals or teams within a given process. In other words, it is quite possible to discover that the difference in performance can be attributed to differences in the personnel executing the same process. For instance, consider a process that requires sales teams to perform a set of activities. A benchmarking exercise identifies best-practice sales teams within the organization. It becomes clear that the best-practice teams are doing something different to perform at a higher level within the same process. So, what can the organization do to improve? Instead of exploring the implementation of new applications or processes, the answer might be as simple as transferring knowledge from best-practice teams to the others.

Appendix 11A: Excel Add-In for Data Envelopment Analysis

This appendix describes an add-in to Microsoft Excel that can be used to perform DEA. The DEA solver consists of two files: dea.xla and deasolve.dll.* Perform the following steps to install the add-in for Microsoft Excel:

Create a new folder on the hard drive. For example, create the new folder DEA inside the Program Files directory.

Download the files dea.xla and deasolve.dll onto the DEA folder that was just created.

* These files can be downloaded from www.crcpress.com/product/isbn/9781439885253

Add dea.xla to Microsoft Excel's Add-ins.

The DEA add-in is now available under the Tools menu (for Excel 97–2003) or in the Add-ins ribbon (Excel 2007 and 2010). The DEA Add-In has two options: New Model and Run Model. The use of the DEA Add-In is illustrated with the data from Example 11.5, shown in Table 11.12. This illustration will start with the creation of a new DEA model. The New Model option of the DEA Add-In opens a dialogue window where the following data must be entered:

- Name of the model
- Number of DMUs (or processes)
- Number of input factors
- Number of output factors

The completed dialogue window for this example appears in Figure 11.5. After OK is pressed on the New Model dialogue window, the DEA Add-In creates two new worksheets: Example.Input and Example.Output. The model data are entered in the corresponding cells for each worksheet. The labels of each table can be modified to fit the description of the current model. Figure 11.6 shows the completed Output sheet for this example. The Input sheet is filled out similarly.

After the Input and Output worksheets have been filled out, the Run Model option of the DEA Add-In can be selected. The Run Model dialogue window appears, displaying the following analysis options:

- *Efficiency*. This output consists of a table displaying the efficiency of each DMU (or process in this case) along with the peer group for relatively inefficient processes. A relatively efficient process has no peer group. By default, this is the only output that the DEA Add-In produces.
- *Best Practice*. This is a table that ranks DMUs according to their average efficiency. It also displays the weight values associated with each input and output. The average efficiency is obtained by applying the weight values to each DMU. The rationale is that the best-practice units are relatively efficient regardless of the set of weights used to measure their performance. The best-practice calculations can be used to detect processes that are relatively efficient only due to an uncharacteristic set of weight values.

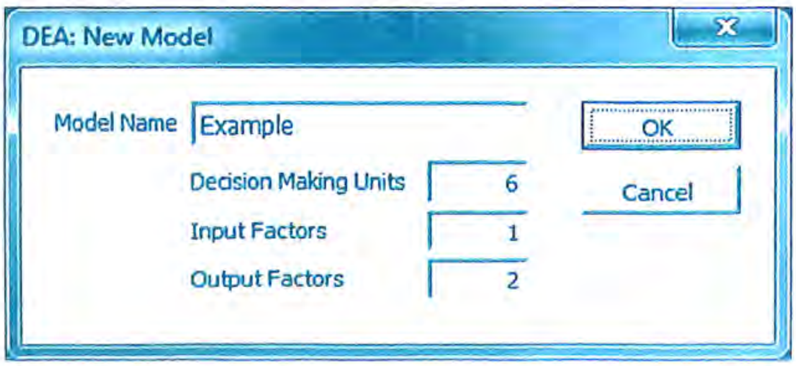
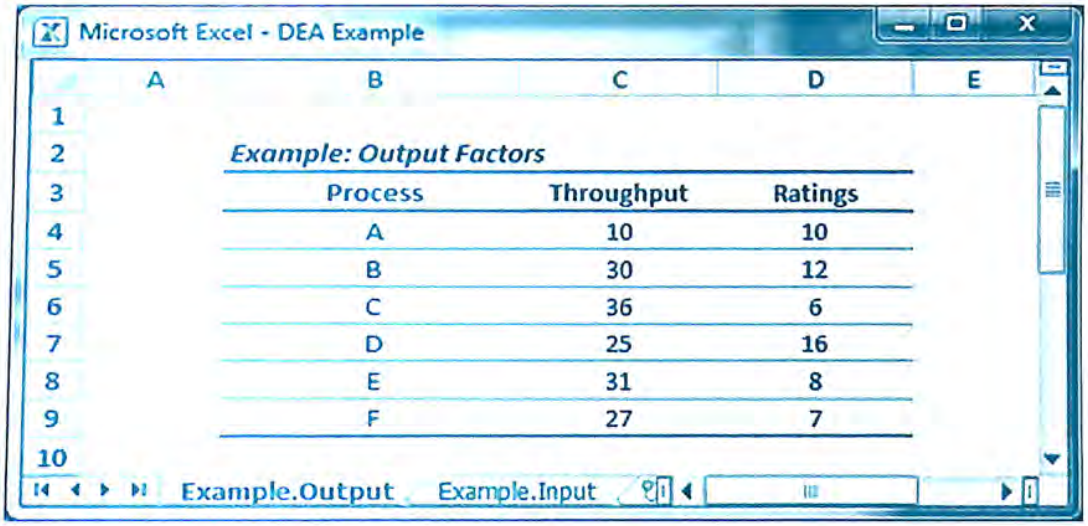


FIGURE 11.5
New model dialogue window.



Example: Output Factors

Process	Throughput	Ratings
A	10	10
B	30	12
C	36	6
D	25	16
E	31	8
F	27	7

FIGURE 11.6
Completed Example.Output worksheet.

- *Targets*. For each relatively inefficient process, this worksheet displays a set of target input and output values that can make the process relatively efficient. As mentioned in Section 11.3, theoretically, an infinite number of target values can turn a relatively inefficient process into a relatively efficient process. In practice, however, certain values cannot be changed easily. For example, if the location of a process is an input in the analysis, changing this value might not be feasible in practice. More sophisticated analysis can be performed to find target values for some inputs or outputs within specified value ranges while keeping values for other inputs and outputs fixed.
- *Virtual Outputs*. This option creates a worksheet and a chart. The worksheet shows the total weighted output for each process. The total weighted output is 100 for relatively efficient processes. The total output for other processes is equal to their efficiency value. The virtual value for Output j and Process k is calculated as the product of the Output j corresponding to Process k from the Output worksheet and the weight for Output j of Process k from the Best Practice worksheet. In the notation introduced in Section 11.3.2, the virtual output is $b_{jk}h_{jk}$. The virtual output chart graphically shows the contribution of each output to the total output of each process. The processes in the chart are ordered by their corresponding efficiency values.
- *Duals**: This information is relevant only to processes that are relatively inefficient. The dual values for relatively efficient processes are 0. For the relatively inefficient processes, the dual values that are not 0 correspond to the constraints associated with a peer process. The dual values are used to create a virtual process for a

* This terminology comes from linear programming, where duality theory roughly establishes that for each primal model with n variables and m constraints, there exists a dual model with m variables and n constraints. The dual variables of the ratio model give the necessary information to create a virtual process for a relatively inefficient process. When the real process is relatively efficient, no virtual process can be more efficient than the real one; therefore, the dual values are equal to 0.

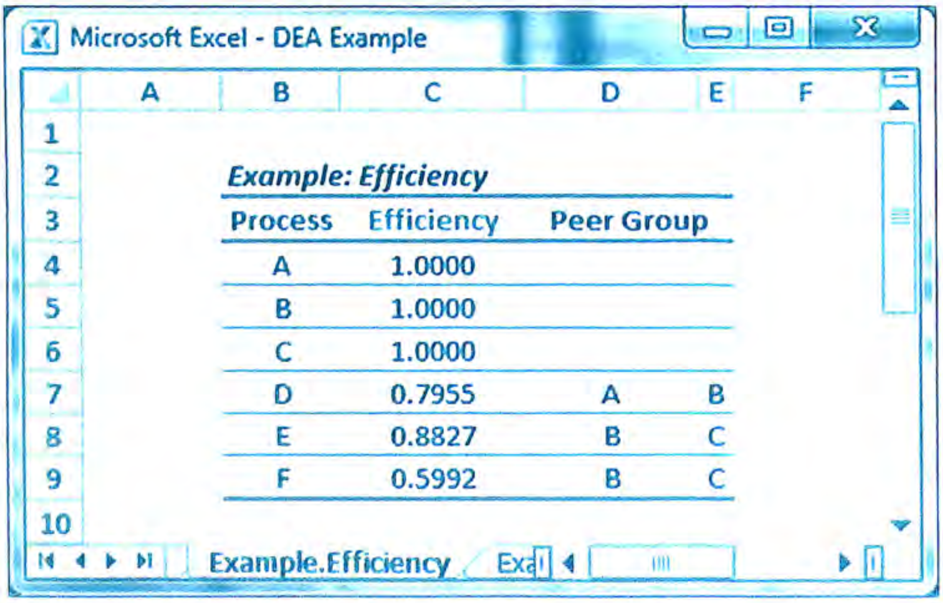
relatively inefficient process. The values in the Target worksheet correspond to the virtual process created with the dual values. This calculation is shown later using the data from Example 11.5.

After all the boxes in the Run Model dialogue window have been checked and OK pressed, the DEA Add-In creates six new worksheets:

- Example.Efficiency
- Example.Best Practice
- Example.Target
- Example.Virtual Outputs
- Example.VO Chart
- Example.Dual

Figure 11.7 shows the table in the Example.Efficiency worksheet. As was shown graphically in Figure 11.3, Processes A, B, and C are relatively efficient, and the other three processes are relatively inefficient. In Section 11.3.1.1, the relative efficiency of Process D was calculated as 0.795, a value that the DEA Add-In finds by solving the linear programming formulation of the ratio model. Figure 11.3 shows that Processes A and B are the peer group of Process D, and the DEA Add-In confirms this finding. Likewise, the peer group for Processes E and F is confirmed as consisting of Processes B and C.

Figure 11.7 shows the table in the Example.Best Practice worksheet. This table shows that Process B is robust in terms of its relative efficiency. Regardless of the set of weights, Process B yields a relative efficiency equal to 1. If the efficiency of Process B is calculated using its chosen weights, 100 percent efficiency is obtained.


Microsoft Excel - DEA Example

Example: Efficiency

Process	Efficiency	Peer Group	
A	1.0000		
B	1.0000		
C	1.0000		
D	0.7955	A	B
E	0.8827	B	C
F	0.5992	B	C

Example.Efficiency

FIGURE 11.7
Example.Efficiency worksheet.

$$\text{Efficiency (Process B)} = \frac{30 \times 1.42857 + 12 \times 4.7619}{15 \times 6.6667} = 1$$

In the same way, it can be easily verified that the efficiency of Process B is still 1 if any of the sets of weights preferred by other processes is used.

Process C has the next-best average efficiency. This process has a relative efficiency of 1 when using its preferred set of weights (8.33333 for labor, 1.78571 for throughput, and 5.95238 for customer ratings). Its average efficiency is not equal to 1, because for some other sets of weights, Process C is not 100 percent efficient. For example, when the set of weights preferred by Process A is applied to Process C, the following efficiency value is obtained:

$$\text{Efficiency (Process C)} = \frac{30 \times 1.66667 + 12 \times 4.33333}{12 \times 10} = 0.9166$$

As shown in Figure 11.8, Process B not only has the best average efficiency but also appears in all the peer groups for relatively inefficient processes. Process C appears in two out of three peer groups, and Process A appears in one.

Figure 11.9 shows the target values for the relatively inefficient processes. These targets are calculated using the dual values in the Example.Dual worksheet. Consider Process D. The peer group for this process consists of Processes A and B. The corresponding dual values for these processes are 1.0 and 0.5 (given in the Example.Dual worksheet not shown here). A weighted combination of Processes A and B creates the virtual process that corresponds to the real Process D, where the weights are the dual values. The input and outputs for the virtual process are as follows:

Target Labor = dual(A) $\times$ labor(A) + dual(B) $\times$ labor(B) = $1 \times 10 + 0.5 \times 15 = 17.5$

Target Throughput = $1 \times 10 + 0.5 \times 30 = 25$

Target Ratings = $1 \times 10 + 0.5 \times 12 = 16$

The resulting target for Process D is the input-oriented target calculated in Section 11.3.1.1.

Microsoft Excel - DEA Example

Example: Best Practice

Process	Average Efficiency	Labor	Throughput	Ratings
B	1.0000	6.66667	1.42857	4.76190
C	0.9722	8.33333	1.78571	5.95238
A	0.9524	10.00000	1.66667	8.33333
E	0.8702	7.14286	1.53061	5.10204
D	0.7738	4.54545	0.75758	3.78788
F	0.5908	5.55556	1.19048	3.96825

Example.Best Practice | Example.Efficiency

FIGURE 11.8
Example.Best practice worksheet.

Microsoft Excel - DEA Example

Example: Targets

Process	Labor	Throughput	Ratings
D	17.5000	25.0000	16.0000
E	12.3571	31.0000	8.0000
F	10.7857	27.0000	7.0000

Example.Target Example.B

FIGURE 11.9
Example.Target worksheet.

Microsoft Excel - DEA Example

Example: Virtual Outputs

Process	Total Output	Throughput	Ratings
B	100.000	42.857	57.143
C	100.000	64.286	35.714
A	100.000	16.667	83.333
E	88.265	47.449	40.816
D	79.545	18.939	60.606
F	59.921	32.143	27.778

Example.Virtual Output Exam

FIGURE 11.10
Example.Virtual outputs worksheet.

Figures 11.10 and 11.11 show the Virtual Output worksheet and the associated chart. The table consists of one column for the total weighted output and one column for the contribution of each output to the total. There is one row for each process in the set. The total weighted output is simply the numerator of the efficiency calculation in the ratio model. In other words, the total output for a process is the sum of the products of each output times its corresponding weight. The output values are given in the Example.Output worksheet, and the weight values are contained in the Example.Best Practice worksheet. For instance, the weighted throughput and ratings values for Process B are calculated as follows:

$$\text{Weighted throughput} = 1.42857 \times 30 = 42.857$$

$$\text{Weighted customer rating} = 4.76190 \times 12 = 57.143$$

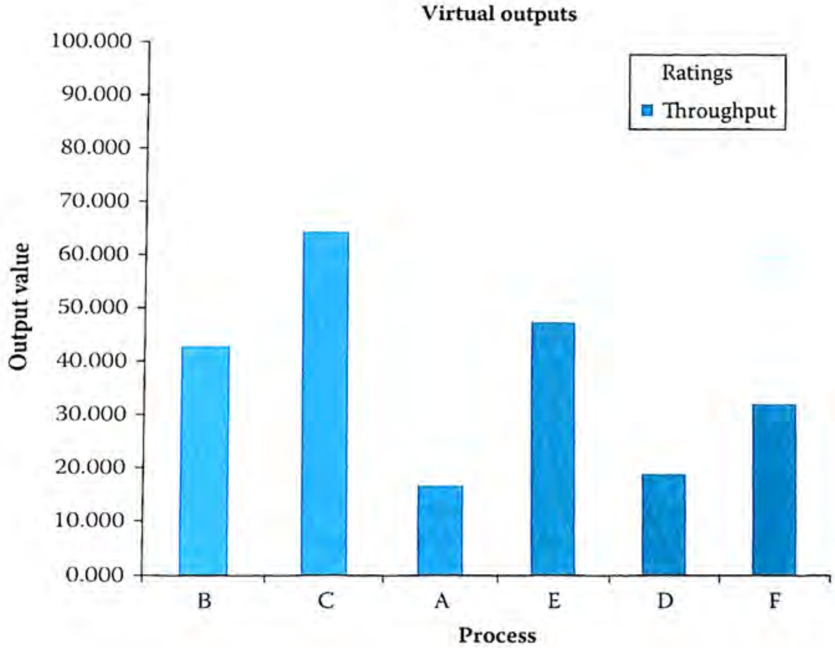


FIGURE 11.11
Virtual output chart.

The bars in the virtual output chart of Figure 11.11 represent the total output for each process. The processes are ordered from maximum to minimum total output. It already has been determined that Process B is not only relatively efficient but also represents the best-practice process. This process has a balanced total output with almost equal contribution from throughput and customer ratings. In contrast, Process A, which is also relatively efficient, obtains most (83.3 percent) of its output from customer ratings. This unbalanced output results in a lower best-practice ranking for Process A.

Discussion Questions and Exercises

1. *Business intelligence and business process management.** Nearly all business processes involve complex value-based decision points, such as loan approvals or up-sell opportunities. Complex decision making may require information that often is not delivered within the workflow, and decision makers must spend time gathering the information required to support the decision. Surveys show that users, on average, spend 20 hours per week gathering and analyzing information. This ad hoc activity may be informal or formal, ranging from accessing and querying a database to soliciting advice from coworkers. Whatever the

* Adapted from "Using business process management and business analytics together for smarter work," IBM Software, WebSphere.

approach, this information-gathering step adds processing time and may create bottlenecks while taking workers away from more productive uses of their time and talent. This leads to inefficient and ineffective processes. Organizations that have invested in business intelligence (BI) solutions to spread the culture of analytics in an organization must make sure that the right information is available to the decision makers in their business processes. The merging of BI and BPM incorporates the right analytic reports from a BI system into a process workflow. The result is that when the workflow gets routed to a decision maker, it is accompanied by relevant and timely information that supports the decision. Give an example of how a business process may benefit from the merger of BI and BPM in the following industries:

a. Banking
b. Healthcare

2. An *if-then* statement is the simplest form of the inference block of a business rule. A set of "if-then" statements is such that the "if" is the condition of the rule, and the "then" is the action. Typically, rules that are based on multiple if-then statements are evaluated sequentially, and each "true" condition triggers some action. Therefore, an if-then statement set may result in more than one action. Consider an online vendor that gives discounts, matches a competitor's price, or offers free shipping to loyal customers depending on the customer loyalty status and the size of the order.
   a. Create an inference block (i.e., a set of if-then statements) for a rule that provides different discounts (5 percent, 10 percent, and 15 percent) to customers according to their loyalty status (bronze, silver, gold).
   b. Enhance the rule to provide free shipping to any loyal customer for orders of more than $100.
   c. Gold customers are given a "best price" guarantee, which means that the price they pay is never higher than the competitors' price. Explain how a business rule could be used to implement this policy.
3. The manager of a commercial loan department for a bank wants to develop a rule to make loan application decisions.* The manager believes that liquidity, profitability, and activity are the three key indicators of performance that are helpful in making a decision on a company's loan application. Liquidity is measured as the ratio of current assets to current liabilities. Profitability is measured as the ratio of net profit to sales. Activity is measured as the ratio of sales to fixed assets. Table 11.14 shows a sample of 20 loans that the bank has made in recent years. The loans are classified into two groups: 1) those that were acceptable and 2) those that should have been rejected.
   a. Use the linear regression approach to develop a classification rule that can be used to accept or reject new loan applications. State the classification rule.
   b. Suppose that five loan applications are received from companies with the key indicators shown in Table 11.15. Use the classification rule to determine which companies should be considered for a loan.

* Adapted from Ragsdale (2011).

TABLE 11.14

Key indicator data for 20 recent loans

Loan	Group	Liquidity	Profitability	Activity
1	1	0.65	0.31	1.75
2	1	0.64	0.29	1.50
3	1	0.62	0.23	1.45
4	1	0.77	0.27	1.20
5	1	0.70	0.28	1.95
6	1	0.85	0.32	1.65
7	1	0.65	0.26	1.79
8	1	0.77	0.29	1.88
9	1	0.64	0.30	1.99
10	1	0.67	0.32	1.84
11	2	0.82	0.25	1.77
12	2	0.65	0.34	1.42
13	2	0.68	0.27	1.91
14	2	0.75	0.22	1.88
15	2	0.85	0.25	1.60
16	2	0.67	0.25	1.34
17	2	0.87	0.22	1.65
18	2	0.82	0.29	1.37
19	2	0.82	0.30	1.46
20	2	0.85	0.29	1.44

TABLE 11.15

Key indicators for five new loan applications

Company	Liquidity	Profitability	Activity
A	0.78	0.27	1.58
B	0.91	0.23	1.67
C	0.68	0.33	1.43
D	0.78	0.23	1.23
E	0.67	0.26	1.78

4. Apply the k-NN classifier to the data in Exercise 3. Use the classifier with $k=3$ to make decisions with respect to the five loan applications in Part (b) of Exercise 3. Are the recommendations from the k-NN classification rule the same as those found with the application of the classification model based on linear regression?
5. Use the graphical approach to DEA and the data in Table 11.16 to determine the efficiency of each process.
6. One major concern with the DEA approach is that with a judicious choice of weights, a high proportion of processes in the set will turn out to be efficient, and DEA will thus have little discriminatory power. Can this happen when a process has the highest ratio of one of the outputs to one of the inputs, considering all the processes in the analysis? Why or why not?
7. Do you think it is possible for a process to appear efficient simply because of its pattern of inputs and outputs and not because of any inherent efficiency? Give a numerical example to illustrate this issue.

TABLE 11.16

Data for Exercise 5

Process	Input	Output 1	Output 2
A	15	90	66
B	20	45	130
C	30	87	147
D	15	75	90
E	30	150	120

8. In some applications of DEA, it has been suggested to impose limits on the weight values for all the processes; that is, the application considers that each weight must be between some specified bounds. Under which circumstances would it be necessary to impose such a range?
9. Consider the linear programming model for Process D shown in Section 11.3.2. The manager of Process D has suggested the use of the following weight values: $h_1 = 0.4$, $h_1 = 0.75$, and $g = 1$ .The manager argues that if each process is allowed to choose weights to maximize its efficiency, he should be allowed to use these values, which clearly show that Process D is relatively efficient. What is wrong with the manager's reasoning?
10. *Warehouse Efficiency.* A distribution system for a large grocery chain consists of 25 warehouses. The director of logistics and transportation would like to evaluate the relative efficiency of each warehouse. Warehouses have a fleet of trucks to deliver grocery items to a set of stores within their region. The director has identified five input factors and four output factors that can be used to evaluate the relative efficiency of the warehouses. The input factors are number of trucks, full-time-equivalent employees, warehouse size, operating expenses, and average number of overtime hours per week. The manager of each warehouse uses overtime hours to pay drivers so that delivery routes can be completed. The output factors are number of deliveries, percentage of on-time deliveries, truck utilization, and customer ratings. Truck utilization is the percentage of time that a truck is actually delivering goods, which excludes traveling time when the truck is empty. This output measure encourages an efficient use of the fleet by employing routes that minimize total travel time. Store managers within each region give customer ratings to their supplying warehouses (10 is a perfect score). Table 11.17 shows data relevant to this analysis.
    a. Use DEA to identify the relatively efficient warehouses.
    b. Find the set of less productive warehouses and identify the percentage of excess resources used by each warehouse in this set.
    c. Identify the reference set (or peer group) for each inefficient warehouse.
    d. The director is concerned with the fact that some warehouses might appear relatively efficient by ignoring within their weighing structure all but very small subsets of their inputs and outputs. The director wants to be sure that relative efficiency is not simply the consequence of a totally unrealistic weighing structure. He would like you to construct a cross-efficiency matrix to determine efficient operating practices. (See Table 11.18.) This matrix conveys information on how a warehouse's relative efficiency is rated by other warehouses.

TABLE 11.17

Data for Exercise 10

No.	No. of trucks	FTE employees	Warehouse size	Operating expenses	Driver overtime	On-time deliveries	Number of deliveries	Truck utilization	Customer rating
1	10	19	9,828	10,995	7	97%	232	0.662	8.10
2	16	25	15,485	14,459	15	84%	370	0.627	6.37
3	11	17	11,625	10,505	8	87%	333	0.872	5.20
4	14	33	13,221	16,750	14	99%	265	0.532	8.69
5	10	21	10,643	10,990	9	81%	211	0.569	9.43
6	19	44	19,312	19,389	16	97%	588	0.937	5.05
7	14	30	13,373	18,410	13	94%	309	0.650	7.71
8	19	31	18,405	17,820	17	90%	420	0.628	9.25
9	20	40	20,782	27,519	16	77%	388	0.539	8.25
10	20	37	19,183	21,748	17	90%	436	0.644	9.38
11	19	40	18,878	20,842	15	75%	568	0.836	5.82
12	16	28	16,661	20,969	16	80%	408	0.708	8.93
13	12	26	12,763	14,226	10	97%	318	0.807	7.79
14	18	34	17,025	17,279	18	89%	441	0.685	5.53
15	11	23	10,178	11,723	10	91%	339	0.859	8.77
16	10	23	9,239	8,673	9	96%	302	0.887	9.57
17	20	49	20,103	25,258	19	82%	575	0.871	6.19
18	18	37	17,262	19,160	14	91%	450	0.744	9.01
19	12	18	12,033	11,131	8	71%	352	0.888	8.39
20	18	39	17,326	23,390	13	90%	532	0.837	7.26
21	14	24	14,555	16,996	13	86%	383	0.821	6.01
22	19	45	18,719	22,281	19	91%	548	0.865	5.51
23	19	31	19,776	20,774	15	78%	480	0.713	5.03
24	11	25	11,690	11,306	7	76%	236	0.578	6.47
25	15	35	14,471	13,861	13	94%	343	0.674	5.31

TABLE 11.18

Efficiency matrix template

	Warehouse		
Warehouse	*1*	2	...
1			
2			
...			
Average			

The entry in Cell (i, j) shows the relative efficiency of Warehouse j with the DEA weights that are optimal for Warehouse i. Then, the average efficiency in each column is computed to get a measure of how the warehouse associated with the column is rated by the rest of the warehouses. A high average efficiency identifies good operating practices. The director believes that this procedure can effectively discriminate between a warehouse that is a self-evaluator and one that is an evaluator of other warehouses.

e. The director also would like to set targets for those warehouses that have been identified as inefficient. Can you recommend some targets for the relatively inefficient warehouses?

f. The director is preparing a presentation to discuss the results of this analysis with the warehouse managers. What output data or exhibits would you recommend the director use for this presentation?

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Epilogue

In the first three chapters, this book examined the issues of process design from a conceptual, high-level view. Chapter 4 began the move from a high-level to a low-level, detailed examination of processes and flows. This detailed examination led readers through deterministic models for cycle time and capacity analysis in Chapter 5, and stochastic queuing models in Chapter 6. Then, Chapter 7 introduced the notion of building computer simulations for modeling business processes. Chapters 8 and 9 moved from theory to practice in the realm of process simulation with the introduction and fairly extensive use of the ExtendSim software. Chapter 10 climbed to a higher-level view of a process by treating simulation models as black boxes. In the black-box view, the internal details of the simulation model were not of concern; instead, the focus was on finding effective values for input parameters, where effectiveness was measured by retrieving relevant output from the simulation model. Finally, Chapter 11 looked at processes from both a white-box and a black-box perspective while embracing analytics as an approach to business process management (BPM). Two particular analytics tools for processes were explained in some detail: data mining and data envelopment analysis. In data mining, the focus is on predictive models that can be employed to construct business rules. Data envelopment analysis, on the other hand, treats a process as a black box and measures effectiveness with a static set of input and output values. The technique, in its simplest form, is not concerned with the dynamics of the process but rather, with the effective transformation of a chosen set of inputs into outputs.

The main goal of this book is to provide a balanced approach to BPM by not only discussing the managerial implications of the approach but also presenting details of the key supporting analytical tools. The quantitative approaches are not meant to be the answer to all problems related to process performance. As discussed in Chapter 7, computer simulation is a very powerful tool; however, it is not appropriate in all situations. The intent has not been to build an argument for the application of quantitative tools and technology but rather, to make the reader familiar with a spectrum of techniques—from the fairly simplistic to the somewhat sophisticated—to support BPM initiatives.

The BPM community is very active both inside and outside academia. Organizations such as the Institute for Operations Research and the Management Sciences (www.informs.org), the Production and Operations Management Society (www.poms.org), and the Institute of Industrial and Systems Engineers (http://www.iise.org) are excellent resources in areas related to analytics and business process excellence within the academic community. Likewise, in the practitioner's world, organizations such as the Association of Business Process Management Professionals (www.abpmp.org) have a wealth of information about the latest developments in the practice of BPM.

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